

AnswerBot- AI-powered FAQ and customer support automation.



Description

The **AI FAQ Assistant** is a robust, secure, and intelligent REST API platform designed to **streamline FAQ creation**, **user authentication**, and **AI-powered content management**.



Built with **Node.js** and **Express.js**, the application utilizes **MongoDB** as its primary data store along with **Mongoose ODM** for elegant data modeling and schema validation.

Security is central to the architecture, implementing **JWT (JSON Web Tokens)** for stateless user sessions and route guarding alongside **bcrypt** for strong password encryption. The backend handles intensive operations like automated FAQ generation, semantic search queries, and dynamic context parsing. It also features robust **centralized error handling** and response status sanitization to prevent unhandled database leakage and ensure operational durability.



Scenario-Based Case Study



Background

Priya is a customer support operations manager who handles massive backlogs of repetitive user inquiries but struggles to organize entries, classify topics, write precise summaries, and track client question trends efficiently.



Problem

- Difficulty generating and maintaining up-to-date documentation workflows.
- Limited visibility into semantic search match keywords and rapid query resolutions.
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- Time-consuming manual processes for drafting complex support answers.
- Lack of unified database permission layers protecting private documentation from public adjustments.



2. Hardware Requirements



Processor

Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent.



RAM

8 GB minimum (16 GB recommended for concurrent instances of MongoDB processing layers).



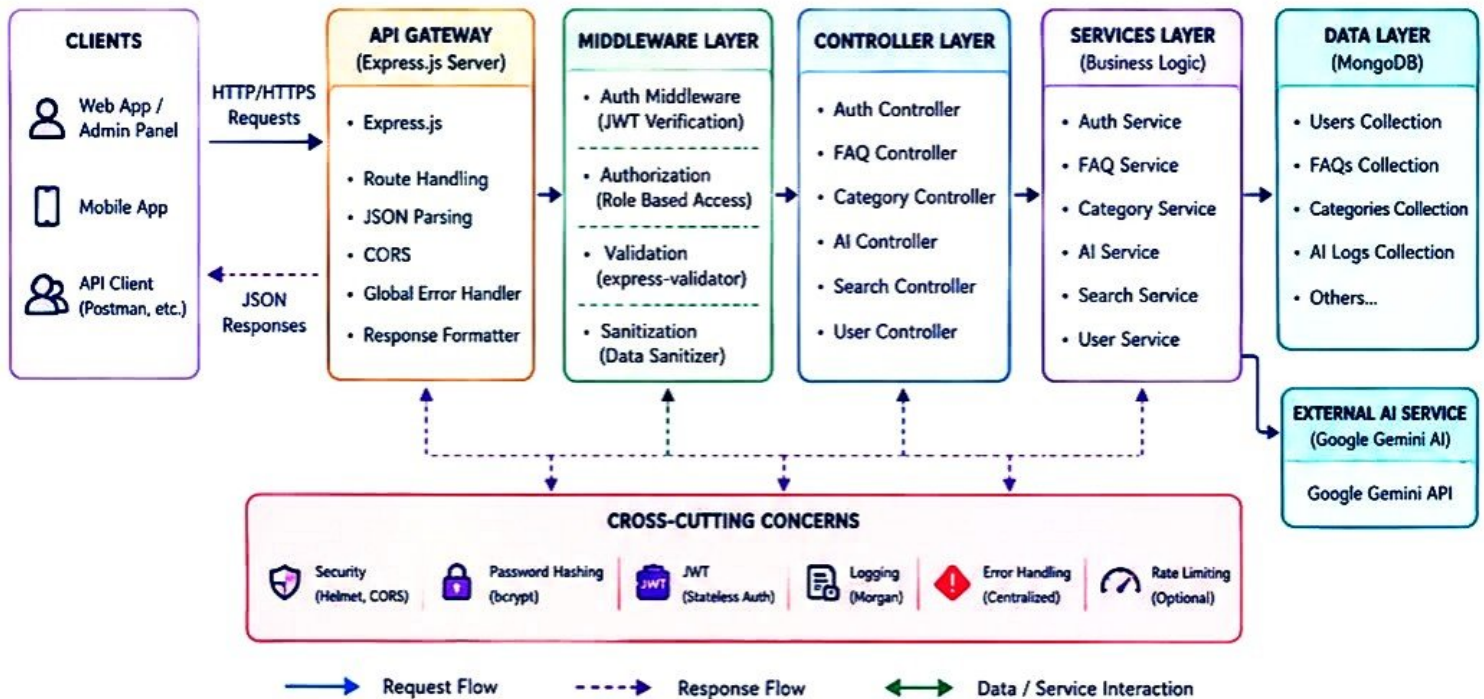
Storage

1 GB of available disk workspace.



Epic-1: Project Architecture

AI FAQ Assistant API – Technical Architecture



The AI FAQ Assistant application exposes structured endpoints handled by a layered server architecture model. Incoming requests undergo filter processing before interacting with data services.



Components Definition



Interface: API tools like Postman query distinct route paths to extract JSON entities.





AI FAQ Assistant Backend & Workflow Overview

Secure • Smart • Scalable



Solution

The AI FAQ Assistant backend delivers a centralized, secure API wrapper that decouples application business logic from frontend delivery systems. It introduces internal verification matrices controlling specialized services for content creation, category keywords, database persistence, and automated text inference modules.



- ✓ Content Creation
- ✓ Category Keywords
- ✓ Database Persistence
- ✓ Text Inference



Usage



Authenticated Creator Usage

Registers profiles, requests instant AI assistance, drafts official FAQ schemas, and updates or removes self-authored records.



Public/Consumer Usage

Queries general list routes to consume published articles and issues localized semantic queries via parameters.



Outcome



By offloading workflow controls to a robust backend, operational delays are minimized.



Data structural integrity is strictly maintained through automated schema validation, security tokens guarantee that only original authors modify documents, and high-performance indexing facilitates real-time data discoverability.



1. Software Requirements



Operating System

Windows 10/11, macOS, or Linux (supports cross-platform operations).



Node.js (v18 or above)

Runtime ecosystem running server logic and routing infrastructure.



npm (v9 or above)

Node package manager to control operational dependencies.



Express.js (v4/v5 Ecosystem)

Lightweight routing web framework to construct backend RESTful entry points.



MongoDB / Mongoose ODM

Document NoSQL engine storing application users and curated data entities.



Postman / ThunderClient

API verification toolkit to assert schema validation responses across administrative route shields.



Code Editor

Visual Studio Code or similar IDE environment.



Key Points

- ✓ API layer decouples frontend and backend logic.
- ✓ Ensures security, data integrity, and real-time searchability.
- ✓ Built for scalable, reliable, and efficient FAQ management.

Key Features & Responsibilities



Category Management: Organizes FAQs into structured categories for easier navigation.



Password Encryption: User passwords are securely hashed using bcrypt before storage.



Centralized Error Handling: Standardized API error responses improve reliability and debugging.



Schema Validation: Mongoose validation ensures data integrity before database persistence.



RESTful API Architecture: Modular Express.js routes provide scalable backend services.



MongoDB Integration: Efficient NoSQL document storage using Mongoose ODM.



Secure • Scalable
Reliable • Fast

Roles and Responsibilities



Admin



- Manage all registered users.
- View, update, and delete any FAQ.
- Monitor AI-generated content.
- Manage FAQ categories.
- Access complete system functionality.



Content Creator



- Create new FAQs.
- Edit personal FAQ records.
- Delete owned FAQs.
- Generate FAQs using AI.
- Organize FAQs under categories.



Authenticated User

- View public FAQs.
- Search FAQs using keywords.
- Generate AI-assisted answers.



Get Answers
Instantly!

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Category Entity (Logical)



- categoryName** (String)
 The name of the category (e.g., Login, Password, Billing).
- description** (String)
 A brief description of the category.
- totalFAQs** (Number)
 Count of FAQs under this category.



Key Structural Mappings



User (1 → Many)

A registered user can create and manage multiple FAQ records.



Category (1 → Many)

Each category can contain multiple FAQ documents.



FAQ (1 → Many)

A user can generate multiple AI-assisted FAQs, which may later be saved into the FAQ collection.



FAQ → Search Results (Logical Relationship)

Search queries retrieve matching FAQ documents using keyword and semantic matching.



Flow: User → Category → FAQ → Stored in DB → Retrieved by Search



Key Features



JWT Authentication & Authorization

- ✓ Secure token-based authentication.
- ✓ Protects private API endpoints.



AI FAQ Generation

- ✓ Google Gemini AI automatically generates questions, answers, and topic suggestions.



Semantic FAQ Search

- ✓ Intelligent keyword-based search.
- ✓ Faster information retrieval.



Complete CRUD Operations

- ✓ Create, Read, Update, and Delete FAQ records.



Smart Automation → Better Support → Happier Users



3. AI Data Model – Entities

Simple • Structured • Connected ♡



1. User Entity

<code>_id</code>	Primary Key – ObjectId
<code>name</code>	String, Required
<code>email</code>	String, Required, Unique
<code>password</code>	String, Hashed, Required
<code>role</code>	String, Default: "user"
<code>createdAt</code>	Date – Automatically added
<code>updatedAt</code>	Date – Automatically added



Stores registered users and their basic details.



Key Points

- ✓ Holds user profiles.
- ✓ Stores hashed passwords for security.
- ✓ Role is "user" by default.
- ✓ Created & updated timestamps are tracked.



2. FAQ Entity

<code>_id</code>	Primary Key – ObjectId
<code>question</code>	String, Required
<code>answer</code>	String, Required
<code>category</code>	String, Required
<code>createdBy</code>	Foreign Key (User Collection)
<code>createdAt</code>	Date – Automatically added
<code>updatedAt</code>	Date – Automatically added



Stores frequently asked questions and their answers.



Key Points

- ✓ Links to the user who created it.
- ✓ Helps users find quick answers.
- ✓ Organized by categories.
- ✓ Supports easy search and retrieval.



3. AI Generation Entity (Logical)

<code>topic</code>	String
<code>generatedQuestion</code>	String
<code>generatedAnswer</code>	String
<code>generatedCategory</code>	String
<code>generatedAt</code>	Date – Automatically added



Stores AI-generated FAQs dynamically (after persistence).



Key Points

- ✓ Generated using Google Gemini AI.
- ✓ Creates questions, answers, and categories automatically.
- ✓ Helps keep the FAQ collection updated and fresh.
- ✓ Works with the same structure as manual FAQs.



From Users → to FAQs → to AI Generated Content
— A smarter way to deliver information — ♡